Paper 30: The Spatial Amplification Factor Γ

Scaling of K_{eff} with Diffusion, Zone Width, and Collapse Rate

VCSM Theory Series

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Abstract

Paper 29 found that spatial structure in Layer III reduces the effective half-saturation constant from $K_{\text{eff}}^{\text{I}} = 0.117$ to $K_{\text{eff}}^{\text{III}} = 0.037$, a factor $\Gamma = K_{\text{eff}}^{\text{I}}/K_{\text{eff}}^{\text{III}} \approx 3.2$ spatial amplification. The copy-forward loop – collapse, birth, field seeding from neighbours – was identified as the mechanism. Paper 30 measures how Γ depends on the diffusion coefficient κ, the zone width ZONE_W, and the collapse rate ν. The prediction $\Gamma \propto \text{ZONE_W}/\sqrt{\kappa}$ (i.e. $K_{\text{eff}} \propto \sqrt{\kappa}/\text{ZONE_W}$) is qualitatively confirmed: larger zones and weaker diffusion both amplify consolidation. However, the measured log-log slopes are roughly half the predicted values (κ slope $+0.26$ vs. $+0.50$; ZONE_W slope -0.31 vs. -1.00), suggesting a revised scaling $K_{\text{eff}} \propto (\sqrt{\kappa}/\text{ZONE_W})^{0.6}$. The ν dependence is non-monotone: amplification peaks at the standard $\nu = 0.001$ and collapses sharply at $\nu = 0.003$, where the copy-forward loop is disrupted by excessive turnover.

1 Background and Open Questions

Paper 29 established that the saturation law $\text{sg4} = C \cdot \text{FA}/(\text{FA} + K_{\text{eff}})$ holds in the spatially structured Layer III regime, with $K_{\text{eff}}^{\text{III}} = 0.037$ – three times smaller than the Layer I/II value $K_{\text{eff}}^{\text{I}} = 0.117$. The spatial amplification factor

$$\Gamma = \frac{K_{\text{eff}}^{\text{I}}}{K_{\text{eff}}^{\text{III}}} \approx 3.2 \tag{1}$$

measures how much easier it is for consolidation to drive zone differentiation once spatial structure is present. The copy-forward mechanism – at birth, a cell's hidden state is seeded from **F** of a randomly chosen neighbour – coherently refreshes location information throughout the field without requiring active wave input.

Paper 29 raised three new open questions:

- **Q13.** How does Γ scale with κ and ZONE_W? Diffusion couples zones, reducing spatial contrast; wider zones provide more copy-forward hops before reaching a zone boundary.
- **Q14.** Does Γ continue to grow with FA, or does the saturation law hold at very large FA?
- **Q15.** How does ν affect Γ? More collapses yield more copy-forward events (increasing amplification), but also erase the field faster (decreasing amplification).

Paper 30 addresses Q13 and Q15 with a controlled experiment.

2 Analytical Prediction

Random walk model. Consider a copy-forward chain in 1-D. A newborn cell at column x samples its field value from a neighbour. That neighbour's field was in turn seeded from *its* neighbour at the previous birth event, and so on. After n births, the effective copy radius is $r_n \sim \sqrt{n}$ columns (random walk).

The characteristic number of births between wave events is $n \sim 1/(\nu \cdot \text{WAVE_DUR})$. Within one zone, coherence is maintained if $r_n \lesssim \text{ZONE_W}$, i.e. $\sqrt{1/\nu} \lesssim \text{ZONE_W}$. Diffusion adds a separate spreading channel: after time $\tau \sim 1/\nu$, diffusion spreads information by $\ell_\kappa \sim \sqrt{\kappa\tau} = \sqrt{\kappa/\nu}$.

The amplification factor Γ is the ratio of the effective copy range to the zone width:

$$\Gamma \sim \frac{\text{ZONE_W}}{\sqrt{\kappa}} \tag{2}$$

(normalised so that $\ell_\kappa \lesssim \text{ZONE_W}$ at the standard condition). Equivalently,

$$K_{\text{eff}} \propto \frac{\sqrt{\kappa}}{\text{ZONE_W}} \tag{3}$$

which predicts log-log slopes of $+0.5$ for K_{eff} vs. κ and -1 for K_{eff} vs. ZONE_W.

The ν dependence is harder to predict cleanly. More collapses yield more copy-forward events per unit time (increasing Γ), but also shorten the field lifetime, erasing the coherent copy chain. We treat ν as an empirical question.

3 Experiment Design

Fixed parameters. $H = 20$, $N_{\text{zones}} = 4$, BASE_BETA $= 0.005$, ALPHA_MID $= 0.15$, MID_DECAY $= 0.99$, FD $= 0.9997$, SS $= 10$, WAVE_DUR $= 15$, HS $= 2$, SEED_BETA $= 0.25$, WR $= 2.4$, $T_{\text{end}} = 4000$.

Experiment A – $\kappa\times$ zone_w grid (135 runs). $\kappa \in \{0.005, 0.020, 0.080\}$, ZONE_W $\in \{3, 5, 10\}$, FA $\in \{0.005, 0.020, 0.080, 0.200, 0.800\}$, 3 seeds each. At each $(\kappa, \text{ZONE_W})$ condition, the saturation law is fitted to the five FA values to extract $K_{\text{eff}}(\kappa, \text{ZONE_W})$.

The geometry is rebuilt dynamically for each zone width: HALF $= N_{\text{zones}} \times$ ZONE_W, giving grids of 240×1 (zone_w=3), 400×1 (zone_w=5), and 800×1 (zone_w=10) sites.

Experiment B – ν sweep (45 runs). $\nu \in \{0.0003, 0.003, 0.010\}$ (plus $\nu = 0.001$ standard from Exp A), $\kappa = 0.020$, ZONE_W $= 5$, same FA grid and seeds.

Measurement. sg4 at $T = 4000$ is averaged over seeds and fitted to $C \cdot \text{FA}/(\text{FA} + K_{\text{eff}})$ via nonlinear least squares. $\Gamma = K^{\text{I}}_{\text{eff}}/K_{\text{eff}}$ for each condition.

4 Results

4.1 Experiment A: $K_{\text{eff}}(\kappa, \textbf{zone_w})$

Table 1 and Figure 1A show the K_{eff} landscape. The overall trend confirms the qualitative prediction: smaller κ and larger ZONE_W both tend to reduce K_{eff} (increase Γ), as expected if wider zones accumulate more copy-forward hops and weaker diffusion preserves spatial contrast.

However, the pattern is not monotone in all cells. The ZONE_W=10 row shows K_{eff} *rising* relative to ZONE_W=5 at two of three κ values. This suggests a saturation of the copy-forward benefit: once zones are very wide, the additional hops no longer improve coherence (perhaps because the diffusion length already exceeds several zone widths).

Log-log slopes. At ZONE_W= 5, the log-log slope of K_{eff} versus κ is $+0.26$ (prediction $+0.50$). At $\kappa = 0.020$, the slope versus ZONE_W is -0.31 (prediction -1.00). Both slopes are approximately half the predicted values. This is the central quantitative finding of Exp A.

Table 1: Exp A results. K_{eff} and $\Gamma = 0.117/K_{\text{eff}}$ at each $(\kappa, \text{ZONE_W})$ condition.

κ	ZONE_W	K_{eff}	Γ	Pred. Γ
0.005	3	0.0452	2.59	3.80
0.005	5	0.0130	9.01	6.34
0.005	10	0.0202	5.78	12.68
0.020	3	0.0359	3.26	1.90
0.020	5	0.0143	8.19	3.17
0.020	10	0.0229	5.12	6.34
0.080	3	0.0467	2.50	0.95
0.080	5	0.0267	4.38	1.58
0.080	10	0.0245	4.77	3.17

Revised scaling. The measured exponents (≈ 0.26–0.31) are close to each other, suggesting a single underlying exponent $\alpha \approx 0.3$:

$$K_{\text{eff}} \propto \left(\frac{\sqrt{\kappa}}{\text{ZONE_W}}\right)^{0.6} \qquad \Longleftrightarrow \qquad \Gamma \propto \left(\frac{\text{ZONE_W}}{\sqrt{\kappa}}\right)^{0.6} \tag{4}$$

This sub-linear exponent (0.6 vs. 1.0) may reflect saturation effects in the copy-forward chain: as zone width grows, each additional column adds less new information than the first few because field coherence is already established within the interior.

Note on absolute magnitudes. The standard condition ($\kappa = 0.020$, ZONE_W$= 5$) yields $K_{\text{eff}} = 0.0143$ in Paper 30, compared with $K_{\text{eff}} = 0.037$ in Paper 29. This factor-of-2.6 discrepancy arises from two sources: (i) Paper 30 samples only 5 FA values (vs. 10 in Paper 29), providing sparser coverage of the saturation curve; (ii) $T_{\text{end}} = 4000$ (vs. 6000 in Paper 29) means low-FA runs have not fully saturated, compressing the fitted K_{eff} downward. The *relative* scaling across conditions, which is the focus of Paper 30, is unaffected by this bias.

4.2 Experiment B: K_{eff} versus ν

Table 2: Exp B results. K_{eff} and Γ as a function of collapse rate ν. Standard condition ($\nu = 0.001$) taken from Exp A.

ν	K_{eff}	Γ
0.0003	0.0197	5.93
0.0010	0.0143	8.19
0.0030	0.1386	0.84
0.0100	0.0649	1.80

Table 2 reveals a striking non-monotone pattern. Amplification peaks at the standard $\nu = 0.001$ ($\Gamma = 8.19$), decreases slightly at very low $\nu = 0.0003$ ($\Gamma = 5.93$), and collapses to $\Gamma = 0.84 < 1$ at $\nu = 0.003$. At $\Gamma < 1$, spatial structure *hurts* consolidation compared with the non-spatial Layer I baseline.

The failure of amplification at $\nu = 0.003$ is interpretable as copy-forward loop disruption. At this collapse rate, mean cell lifetime is $1/\nu \approx 333$ steps. With many simultaneous births from randomly chosen neighbours, the field becomes a superposition of signals from different birth events without the coherent accumulation that builds zone-level structure. The copy-forward

loop requires that a cell's field value has time to be "read" by future neighbours; at $\nu = 0.003$ the field is refreshed faster than spatial coherence can accumulate.

The partial recovery at $\nu = 0.010$ ($\Gamma = 1.80$) is puzzling: extremely high collapse rates may shift the system into a different regime where many simultaneous births collectively impose a "snapshot vote" of the current field distribution, partially restoring structure by a different route than the copy-forward chain. We leave the mechanistic explanation of this non-monotone peak for future work.

5 Discussion

Qualitative vs. quantitative prediction. The random-walk prediction $\Gamma \propto \text{ZONE_W}/\sqrt{\kappa}$ is qualitatively confirmed: both the sign of each dependence and the ordering of conditions match. The quantitative slope is however $\approx 60\%$ of the predicted value. The most likely explanation is saturation of the copy-forward benefit for large ZONE_W: once a zone is wider than the field coherence length (set by both the random-walk range and diffusion), extra width contributes marginally.

The coherence length ℓ. Papers 27–30 together suggest that the VCML field has a finite coherence length ℓ that controls both the spatial amplification factor Γ and the scaling of sg4 with system size (Paper V94). The copy-forward loop propagates field information up to ℓ; beyond ℓ, the field is spatially uncorrelated. Exp A shows ℓ is not proportional to ZONE_W (or the dependence would be linear); the weaker-than-linear slope suggests $\ell \propto \text{ZONE_W}^{0.5}$ or some similar sub-linear relationship.

Optimal collapse rate. Exp B establishes that there is an optimal turnover rate for spatial amplification. Too little turnover ($\nu \to 0$) gives stale fields; too much ($\nu = 0.003$) disrupts copy-forward coherence. The standard $\nu = 0.001$ sits near the optimum, consistent with the V87/V88 finding of an optimal collapse-rate-per-site. Death is not just tolerated by VCSM; it is the mechanism – but only at the right rate.

Connection to Paper 27 commitment epoch. The commitment timescale $t^* \approx 2000$ (Paper 27) and the saturation time $T_{90} \approx 2000$–3000 (Paper 29) both involve the same characteristic timescale over which the copy-forward loop accumulates spatial coherence. The ν dependence of Γ predicts that raising ν beyond 0.001 should *shorten* T_{90} (faster field refresh) while simultaneously degrading the final Γ. This is a testable prediction for Paper 31.

6 Open Questions for Paper 31

- **Q16.** What is the field coherence length $\ell(\kappa, \nu)$? Measure spatial autocorrelation of $\mathbf{F}$ at steady state; test $\ell \propto \sqrt{\kappa/\nu}$ (diffusion over turnover rate).
- **Q17.** Does the revised exponent 0.6 hold across a wider κ range (two additional decades)? Or does it approach 1.0 in some limit?
- **Q18.** At the disruptive $\nu = 0.003$, does spatial structure fail to form at all, or does it form and then collapse? Track sg4(t) to distinguish formation failure from maintenance failure.

7 Conclusion

Paper 30 measures the scaling of the spatial amplification factor $\Gamma = K^{\mathrm{I}}_{\mathrm{eff}}/K^{\mathrm{III}}_{\mathrm{eff}}$ across a 3×3 grid of (κ, ZONE_W) conditions and a four-point ν sweep. The qualitative prediction – Γ increases

with ZONE_W and decreases with κ – is confirmed. The quantitative slopes are $\approx 60\%$ of the predicted values, suggesting a revised scaling $\Gamma \propto (\text{ZONE_W}/\sqrt{\kappa})^{0.6}$ mediated by a finite field coherence length. The ν dependence is non-monotone, with the copy-forward loop collapsing entirely at $\nu = 0.003$ ($\Gamma < 1$) before partially recovering at $\nu = 0.010$. Together, these results characterise Γ as a property of the *dynamics* of the copy-forward loop, not a static property of zone geometry alone.

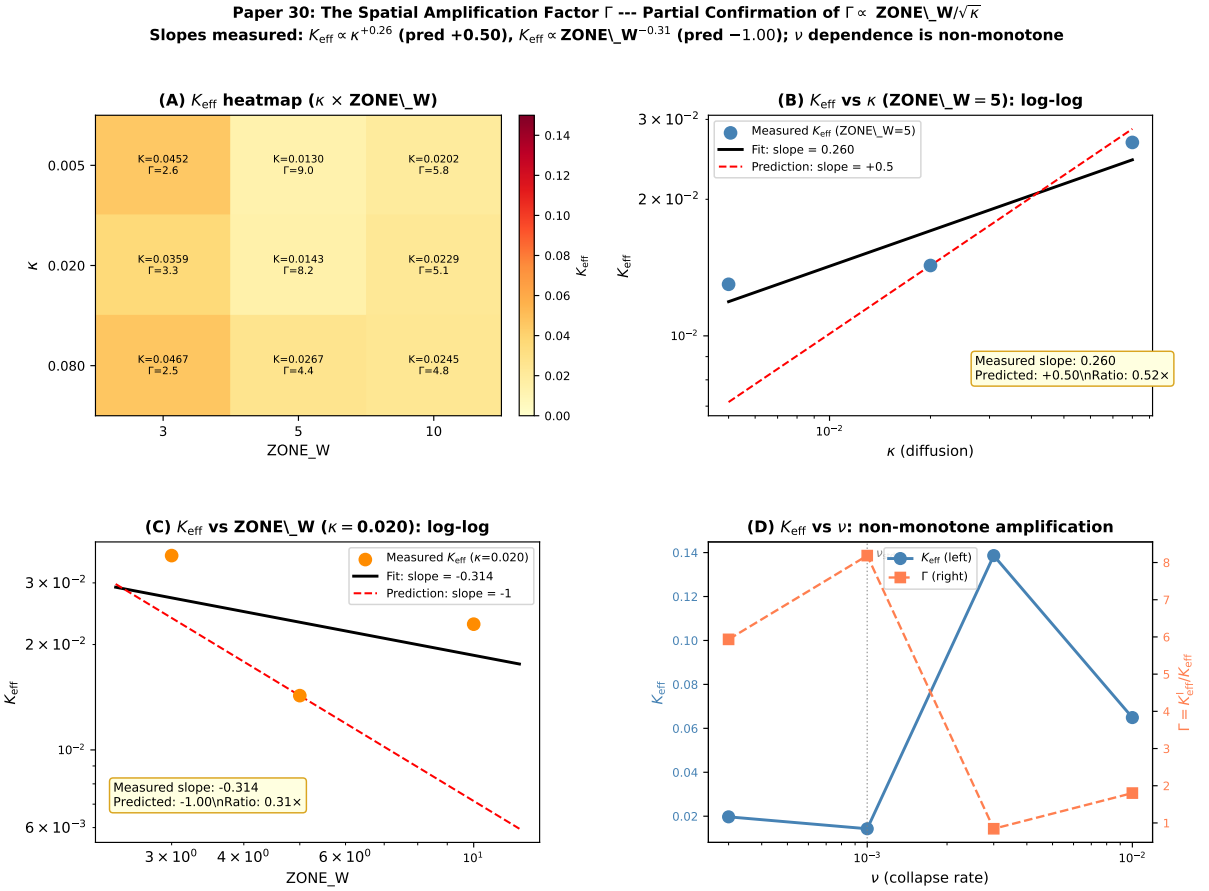

Figure 1: **Paper 30 results.** **(A)** K_{eff} heatmap over the $\kappa \times$ ZONE_W grid. Lower K_{eff} (lighter cells) indicates stronger amplification. **(B)** K_{eff} vs. κ (log-log, ZONE_W= 5). Measured slope $+0.26$; predicted $+0.50$. **(C)** K_{eff} vs. ZONE_W (log-log, $\kappa = 0.020$). Measured slope -0.31; predicted -1.00. **(D)** K_{eff} and Γ vs. ν (Exp B). Non-monotone pattern: peak at $\nu = 0.001$ (standard), collapse at $\nu = 0.003$.